CYBERSRS

Firewall Management System Requirements

Software Requirements Specification

DRAFT — human review required

Version 1 • 2026-10-29T20:54:03.649377+00:00

AI-generated advisory document. Review by a qualified professional is required.

Automated checks do not prove completeness or semantic correctness. SDLC suitability scores are rule-based points, not probabilities.

Content SHA-256: cb1c2569aac5f09c36d3288fa25ea70283b932044fba77a781463d1625989660

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1. Introduction

Original project description

Design a centralized firewall management system for a university that manages firewall policies across academic, administrative, research, hostel, and guest networks. The system should allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts. The platform should maintain audit logs of firewall policy changes and administrative actions, provide a dashboard for reviewing network security events, and support multiple campus buildings. The solution should be secure, scalable, reliable, and manageable.

Summary: This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

Items

Stakeholder source statement E-001

Detail: Design a centralized firewall management system for a university that manages firewall policies across academic, administrative, research, hostel, and guest networks.

Status: confirmed

Evidence ids: E-001

Stakeholder source statement E-002

Detail: The system should allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts.

Status: confirmed

Evidence ids: E-002

Stakeholder source statement E-003

Detail: The platform should maintain audit logs of firewall policy changes and administrative actions, provide a dashboard for reviewing network security events, and support multiple campus buildings.

Status: confirmed

Evidence ids: E-003

Stakeholder source statement E-004

Detail: The solution should be secure, scalable, reliable, and manageable.

Status: confirmed

Evidence ids: E-004

2. Functional requirements

Applicability reason: Independent source obligations; unspecified choices remain TBD.

Requirements

FR-001 - manage firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system

Statement: The system shall manage firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system.

Rationale: Source obligation E-001.

Priority: must

Dimensions: business; security

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: the system manages firewall policies across academic, administrative, research, hostel, and guest networks

When: a policy is updated via the centralized management interface

Then: the change is reflected across all network segments without service disruption

Method: test

Observable evidence: External test capture of the resulting state

FR-002 - allow authorized network administrators to create firewall rules

Statement: The system shall allow authorized network administrators to create firewall rules.

Rationale: Source obligation E-002_O01.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: authorized network administrators create firewall rules

When: the actor initiates the creation of new firewall rules

Then: the system returns a newly created rule with specified parameters and status

Method: test

Observable evidence: External test capture of the resulting response

FR-003 - allow authorized network administrators to review firewall rules

Statement: The system shall allow authorized network administrators to review firewall rules.

Rationale: Source obligation E-002_O02.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: authorized network administrators request to review firewall rules

When: the actor initiates a rule review operation

Then: the requested firewall rules are returned in full detail for examination

Method: inspection

Observable evidence: External test capture of the resulting response

FR-004 - allow authorized network administrators to update firewall rules

Statement: The system shall allow authorized network administrators to update firewall rules.

Rationale: Source obligation E-002_O03.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: authorized network administrators update existing firewall rules

When: the actor initiates an update to a specific rule

Then: the system applies the updated rule and confirms its new state in the configuration

Method: test

Observable evidence: External test capture of the resulting state

FR-005 - allow authorized network administrators to revoke firewall rules

Statement: The system shall allow authorized network administrators to revoke firewall rules.

Rationale: Source obligation E-002_O04.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: authorized network administrators revoke firewall rules

When: the actor initiates a rule revocation action

Then: the system removes the rule from active configuration and marks it as revoked

Method: test

Observable evidence: External test capture of the resulting state

FR-006 - enforce access restrictions between network segments

Statement: The system shall enforce access restrictions between network segments.

Rationale: Source obligation E-002_O05.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: the system enforces access restrictions between network segments

When: a communication attempt is made between isolated segments

Then: the system blocks the connection.

Method: demonstration

Observable evidence: External test capture of the resulting state

FR-007 - monitor blocked and allowed traffic

Statement: The system shall monitor blocked and allowed traffic.

Rationale: Source obligation E-002_O06.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: the system monitors traffic between network segments

When: traffic flows through the firewall interface

Then: The system reports an observable result showing traffic flow states between network segments as monitored through the firewall interface.

Method: analysis

Observable evidence: External test capture of the resulting state

FR-008 - generate alerts for suspicious or unauthorized communication attempts

Statement: The system shall generate alerts for suspicious or unauthorized communication attempts.

Rationale: Source obligation E-002_O07.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: a suspicious communication attempt is detected

When: an unauthorized or anomalous connection is initiated

Then: the system generates an alert indicating the event and its context

Method: demonstration

Observable evidence: External test capture of the resulting response

FR-009 - maintain audit logs of firewall policy changes and administrative actions

Statement: The system shall maintain audit logs of firewall policy changes and administrative actions.

Rationale: Source obligation E-003_O01.

Priority: should

Dimensions: business; security; auditability; data

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: system maintains audit logs of firewall policy changes and administrative actions

When: the system records a policy change or administrative action

Then: The system records a policy change or administrative action and the external test captures the resulting response.

Method: test

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-3aae42e94793 is TBD.; Decision Q-64ef2e28335c is TBD.

FR-010 - provide a dashboard for reviewing network security events

Statement: The system shall provide a dashboard for reviewing network security events.

Rationale: Source obligation E-003_O02.

Priority: should

Dimensions: business; data

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: system provides a dashboard for reviewing network security events

When: an administrator accesses the dashboard interface

Then: the dashboard displays security event summaries with filtering and sorting capabilities

Method: demonstration

Observable evidence: External test capture of the resulting state

FR-011 - support multiple campus buildings

Statement: The system shall support multiple campus buildings.

Rationale: Source obligation E-003_O03.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: system supports multiple campus buildings

When: an administrative user initiates a network configuration request from each building's designated location

Then: the request is processed successfully in each location with no error or timeout

Method: test

Observable evidence: External test capture of the resulting response

3. Non-functional requirements

Applicability reason: Independent source obligations; unspecified choices remain TBD.

Requirements

NFR-001 - be secure

Statement: The system shall be secure.

Rationale: Source obligation E-004_O01.

Priority: should

Dimensions: security

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: A draft acceptance plan for the system to be secure

When: An analyst reviews that acceptance plan

Then: The plan's measurable targets remain TBD or unconfirmed

Method: inspection

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-7b2e5f89a0b1 is TBD.

NFR-002 - be scalable

Statement: The system shall be scalable.

Rationale: Source obligation E-004_O02.

Priority: should

Dimensions: performance

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: A draft acceptance plan for the system to be scalable

When: An analyst reviews that acceptance plan

Then: The plan's measurable targets remain TBD or unconfirmed

Method: inspection

Observable evidence: External test capture of the resulting state

Unresolved decisions: Decision Q-543394346815 is TBD.

NFR-003 - be reliable

Statement: The system shall be reliable.

Rationale: Source obligation E-004_O03.

Priority: should

Dimensions: availability

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: A draft acceptance plan for the system to be reliable

When: An analyst reviews that acceptance plan

Then: The plan's measurable targets remain TBD or unconfirmed

Method: inspection

Observable evidence: External test capture of the inspection record

Unresolved decisions: Decision Q-819bb986f419 is TBD.

NFR-004 - be manageable

Statement: The system shall be manageable.

Rationale: Source obligation E-004_O04.

Priority: should

Dimensions: usability

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: A draft acceptance plan for the system to be manageable

When: An analyst reviews that acceptance plan

Then: The plan's measurable targets remain TBD or unconfirmed

Method: inspection

Observable evidence: External test capture of the inspection record

Unresolved decisions: Decision Q-5268b96e41b0 is TBD.

4. Security, privacy and compliance

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; NFR-001

5. Data requirements

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-009; FR-010; FR-011

6. Network and interfaces

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-009; FR-010

7. Compliance and policy applicability matrix

8. Architecture

Summary: Proposed logical components and interactions mapped to canonical source obligations.

Items

Actor: authorized network administrators

Detail: Proposed logical endpoint for the source-stated actor: authorized network administrators.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002; FR-003; FR-004; FR-005

Component name: Actor: authorized network administrators

Responsibility: Represent the source-stated actor: authorized network administrators.

Firewall Policy Management Engine

Detail: Proposed logical owner for FR-001, FR-006, FR-011, NFR-001, NFR-002, NFR-003, NFR-004.

Status: proposed

Evidence ids: E-001; E-002; E-003; E-004

Requirement ids: FR-001; FR-006; FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Component name: Firewall Policy Management Engine

Responsibility: The system shall manage firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system. The system shall enforce access restrictions between network segments. The system shall support multiple campus buildings. The system shall be secure. The system shall be scalable. The system shall be reliable. The system shall be manageable.

Rule Administration Interface

Detail: Proposed logical owner for FR-002, FR-003, FR-004, FR-005, FR-011, NFR-001, NFR-002, NFR-003, NFR-004.

Status: proposed

Evidence ids: E-002; E-003; E-004

Requirement ids: FR-002; FR-003; FR-004; FR-005; FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Component name: Rule Administration Interface

Responsibility: The system shall allow authorized network administrators to create firewall rules. The system shall allow authorized network administrators to review firewall rules. The system shall allow authorized network administrators to update firewall rules. The system shall allow authorized network administrators to revoke firewall rules. The system shall support multiple campus buildings. The system shall be secure. The system shall be scalable. The system shall be reliable. The system shall be manageable.

Traffic Monitoring and Alerting Module

Detail: Proposed logical owner for FR-007, FR-008, FR-011, NFR-001, NFR-002, NFR-003, NFR-004.

Status: proposed

Evidence ids: E-002; E-003; E-004

Requirement ids: FR-007; FR-008; FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Component name: Traffic Monitoring and Alerting Module

Responsibility: The system shall monitor blocked and allowed traffic. The system shall generate alerts for suspicious or unauthorized communication attempts. The system shall support multiple campus buildings. The system shall be secure. The system shall be scalable. The system shall be reliable. The system shall be manageable.

Audit Logging and Reporting Service

Detail: Proposed logical owner for FR-009, FR-010, FR-011, NFR-001, NFR-002, NFR-003, NFR-004.

Status: proposed

Evidence ids: E-003; E-004

Requirement ids: FR-009; FR-010; FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Component name: Audit Logging and Reporting Service

Responsibility: The system shall maintain audit logs of firewall policy changes and administrative actions. The system shall provide a dashboard for reviewing network security events. The system shall support multiple campus buildings. The system shall be secure. The system shall be scalable. The system shall be reliable. The system shall be manageable.

Data flows

Proposed source-defined interaction

Detail: Proposed directed path for FR-002, FR-003, FR-004, FR-005.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002; FR-003; FR-004; FR-005

From component: Actor: authorized network administrators

To component: Rule Administration Interface

Data or action: The system shall allow authorized network administrators to create firewall rules. The system shall allow authorized network administrators to review firewall rules. The system shall allow authorized network administrators to update firewall rules. The system shall allow authorized network administrators to revoke firewall rules.

Trust boundaries

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-002, FR-003, FR-004, FR-005.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002; FR-003; FR-004; FR-005

Untrusted component: Actor: authorized network administrators

Trusted component: Rule Administration Interface

Enforcement: The system shall allow authorized network administrators to create firewall rules. The system shall allow authorized network administrators to review firewall rules. The system shall allow authorized network administrators to update firewall rules. The system shall allow authorized network administrators to revoke firewall rules.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-011, NFR-001, NFR-002, NFR-003, NFR-004.

Status: proposed

Evidence ids: E-003; E-004

Requirement ids: FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Untrusted component: Firewall Policy Management Engine

Trusted component: Rule Administration Interface

Enforcement: The system shall support multiple campus buildings. The system shall be secure. The system shall be scalable. The system shall be reliable. The system shall be manageable.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-011, NFR-001, NFR-002, NFR-003, NFR-004.

Status: proposed

Evidence ids: E-003; E-004

Requirement ids: FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Untrusted component: Firewall Policy Management Engine

Trusted component: Traffic Monitoring and Alerting Module

Enforcement: The system shall support multiple campus buildings. The system shall be secure. The system shall be scalable. The system shall be reliable. The system shall be manageable.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-011, NFR-001, NFR-002, NFR-003, NFR-004.

Status: proposed

Evidence ids: E-003; E-004

Requirement ids: FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Untrusted component: Firewall Policy Management Engine

Trusted component: Audit Logging and Reporting Service

Enforcement: The system shall support multiple campus buildings. The system shall be secure. The system shall be scalable. The system shall be reliable. The system shall be manageable.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-011, NFR-001, NFR-002, NFR-003, NFR-004.

Status: proposed

Evidence ids: E-003; E-004

Requirement ids: FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Untrusted component: Rule Administration Interface

Trusted component: Firewall Policy Management Engine

Enforcement: The system shall support multiple campus buildings. The system shall be secure. The system shall be scalable. The system shall be reliable. The system shall be manageable.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-011, NFR-001, NFR-002, NFR-003, NFR-004.

Status: proposed

Evidence ids: E-003; E-004

Requirement ids: FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Untrusted component: Rule Administration Interface

Trusted component: Traffic Monitoring and Alerting Module

Enforcement: The system shall support multiple campus buildings. The system shall be secure. The system shall be scalable. The system shall be reliable. The system shall be manageable.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-011, NFR-001, NFR-002, NFR-003, NFR-004.

Status: proposed

Evidence ids: E-003; E-004

Requirement ids: FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Untrusted component: Rule Administration Interface

Trusted component: Audit Logging and Reporting Service

Enforcement: The system shall support multiple campus buildings. The system shall be secure. The system shall be scalable. The system shall be reliable. The system shall be manageable.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-011, NFR-001, NFR-002, NFR-003, NFR-004.

Status: proposed

Evidence ids: E-003; E-004

Requirement ids: FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Untrusted component: Traffic Monitoring and Alerting Module

Trusted component: Firewall Policy Management Engine

Enforcement: The system shall support multiple campus buildings. The system shall be secure. The system shall be scalable. The system shall be reliable. The system shall be manageable.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-011, NFR-001, NFR-002, NFR-003, NFR-004.

Status: proposed

Evidence ids: E-003; E-004

Requirement ids: FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Untrusted component: Traffic Monitoring and Alerting Module

Trusted component: Rule Administration Interface

Enforcement: The system shall support multiple campus buildings. The system shall be secure. The system shall be scalable. The system shall be reliable. The system shall be manageable.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-011, NFR-001, NFR-002, NFR-003, NFR-004.

Status: proposed

Evidence ids: E-003; E-004

Requirement ids: FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Untrusted component: Traffic Monitoring and Alerting Module

Trusted component: Audit Logging and Reporting Service

Enforcement: The system shall support multiple campus buildings. The system shall be secure. The system shall be scalable. The system shall be reliable. The system shall be manageable.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-011, NFR-001, NFR-002, NFR-003, NFR-004.

Status: proposed

Evidence ids: E-003; E-004

Requirement ids: FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Untrusted component: Audit Logging and Reporting Service

Trusted component: Firewall Policy Management Engine

Enforcement: The system shall support multiple campus buildings. The system shall be secure. The system shall be scalable. The system shall be reliable. The system shall be manageable.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-011, NFR-001, NFR-002, NFR-003, NFR-004.

Status: proposed

Evidence ids: E-003; E-004

Requirement ids: FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Untrusted component: Audit Logging and Reporting Service

Trusted component: Rule Administration Interface

Enforcement: The system shall support multiple campus buildings. The system shall be secure. The system shall be scalable. The system shall be reliable. The system shall be manageable.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-011, NFR-001, NFR-002, NFR-003, NFR-004.

Status: proposed

Evidence ids: E-003; E-004

Requirement ids: FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Untrusted component: Audit Logging and Reporting Service

Trusted component: Traffic Monitoring and Alerting Module

Enforcement: The system shall support multiple campus buildings. The system shall be secure. The system shall be scalable. The system shall be reliable. The system shall be manageable.

9. Threats

Assumptions

1. These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

10. Risks

11. User stories and use cases

User stories and use cases

Id: UC-001

Actor: authorized network administrators

Goal: The system shall allow authorized network administrators to create firewall rules.

Story: As authorized network administrators, I need the following behavior: The system shall allow authorized network administrators to create firewall rules.

Preconditions

1. authorized network administrators create firewall rules

Steps

1. the actor initiates the creation of new firewall rules

Success outcome: the system returns a newly created rule with specified parameters and status

Requirement ids: FR-002

Evidence ids: E-002

Id: UC-002

Actor: authorized network administrators

Goal: The system shall allow authorized network administrators to review firewall rules.

Story: As authorized network administrators, I need the following behavior: The system shall allow authorized network administrators to review firewall rules.

Preconditions

1. authorized network administrators request to review firewall rules

Steps

1. the actor initiates a rule review operation

Success outcome: the requested firewall rules are returned in full detail for examination

Requirement ids: FR-003

Evidence ids: E-002

Id: UC-003

Actor: authorized network administrators

Goal: The system shall allow authorized network administrators to update firewall rules.

Story: As authorized network administrators, I need the following behavior: The system shall allow authorized network administrators to update firewall rules.

Preconditions

1. authorized network administrators update existing firewall rules

Steps

1. the actor initiates an update to a specific rule

Success outcome: the system applies the updated rule and confirms its new state in the configuration

Requirement ids: FR-004

Evidence ids: E-002

Id: UC-004

Actor: authorized network administrators

Goal: The system shall allow authorized network administrators to revoke firewall rules.

Story: As authorized network administrators, I need the following behavior: The system shall allow authorized network administrators to revoke firewall rules.

Preconditions

1. authorized network administrators revoke firewall rules

Steps

1. the actor initiates a rule revocation action

Success outcome: the system removes the rule from active configuration and marks it as revoked

Requirement ids: FR-005

Evidence ids: E-002

12. Verification and acceptance

Tests

Id: TEST-001

Requirement ids: FR-001

Procedure

1. Given: the system manages firewall policies across academic, administrative, research, hostel, and guest networks
2. When: a policy is updated via the centralized management interface

Expected result: Expected: the change is reflected across all network segments without service disruption

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-002

Requirement ids: FR-002

Procedure

1. Given: authorized network administrators create firewall rules
2. When: the actor initiates the creation of new firewall rules

Expected result: Expected: the system returns a newly created rule with specified parameters and status

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-003

Requirement ids: FR-003

Procedure

1. Given: authorized network administrators request to review firewall rules
2. When: the actor initiates a rule review operation

Expected result: Expected: the requested firewall rules are returned in full detail for examination

Evidence to record: Evidence: External test capture of the resulting response

Test type: inspection

Id: TEST-004

Requirement ids: FR-004

Procedure

1. Given: authorized network administrators update existing firewall rules
2. When: the actor initiates an update to a specific rule

Expected result: Expected: the system applies the updated rule and confirms its new state in the configuration

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-005

Requirement ids: FR-005

Procedure

1. Given: authorized network administrators revoke firewall rules
2. When: the actor initiates a rule revocation action

Expected result: Expected: the system removes the rule from active configuration and marks it as revoked

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-006

Requirement ids: FR-006

Procedure

1. Given: the system enforces access restrictions between network segments
2. When: a communication attempt is made between isolated segments

Expected result: Expected: the system blocks the connection.

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-007

Requirement ids: FR-007

Procedure

1. Given: the system monitors traffic between network segments
2. When: traffic flows through the firewall interface

Expected result: Expected: The system reports an observable result showing traffic flow states between network segments as monitored through the firewall interface.

Evidence to record: Evidence: External test capture of the resulting state

Test type: analysis

Id: TEST-008

Requirement ids: FR-008

Procedure

1. Given: a suspicious communication attempt is detected
2. When: an unauthorized or anomalous connection is initiated

Expected result: Expected: the system generates an alert indicating the event and its context

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-009

Requirement ids: FR-009

Procedure

1. Given: system maintains audit logs of firewall policy changes and administrative actions
2. When: the system records a policy change or administrative action

Expected result: Expected: The system records a policy change or administrative action and the external test captures the resulting response.

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-010

Requirement ids: FR-010

Procedure

1. Given: system provides a dashboard for reviewing network security events
2. When: an administrator accesses the dashboard interface

Expected result: Expected: the dashboard displays security event summaries with filtering and sorting capabilities

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-011

Requirement ids: FR-011

Procedure

1. Given: system supports multiple campus buildings
2. When: an administrative user initiates a network configuration request from each building's designated location

Expected result: Expected: the request is processed successfully in each location with no error or timeout

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-012

Requirement ids: NFR-001

Procedure

1. Given: A draft acceptance plan for the system to be secure
2. When: An analyst reviews that acceptance plan

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed

Evidence to record: Evidence: External test capture of the resulting response

Test type: inspection

Id: TEST-013

Requirement ids: NFR-002

Procedure

1. Given: A draft acceptance plan for the system to be scalable
2. When: An analyst reviews that acceptance plan

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed

Evidence to record: Evidence: External test capture of the resulting state

Test type: inspection

Id: TEST-014

Requirement ids: NFR-003

Procedure

1. Given: A draft acceptance plan for the system to be reliable
2. When: An analyst reviews that acceptance plan

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed

Evidence to record: Evidence: External test capture of the inspection record

Test type: inspection

Id: TEST-015

Requirement ids: NFR-004

Procedure

1. Given: A draft acceptance plan for the system to be manageable
2. When: An analyst reviews that acceptance plan

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed

Evidence to record: Evidence: External test capture of the inspection record

Test type: inspection

Limitations: These verification plans have not been executed; their acceptance outcomes require testing.

13. Glossary

Summary: Source-present terminology; definitions introduce no additional product obligation.

Items

Policy

Detail: A rule or decision governing permitted actions; its project-specific meaning remains defined by the source requirements.

Status: proposed

Evidence ids: E-001; E-003

Audit record

Detail: A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

Status: proposed

Evidence ids: E-003

Log

Detail: A collection of event records. This term does not itself require additional logging fields, events or access permissions.

Status: proposed

Evidence ids: E-003

Firewall

Detail: A control that permits or restricts network traffic according to rules; mentioning it does not authorize automatic changes.

Status: proposed

Evidence ids: E-001; E-002; E-003

14. SDLC recommendation and workflow

Selected model: Agile

Justification: The selection of Agile as the required model is based on a uniform score of 40 across all candidates, with no differentiating input evidence provided to favor one methodology over another. This lack of distinguishing factors—such as budget constraints, failure consequences, regulatory demands, or feedback cycles—means that no candidate model can be objectively ranked for suitability beyond equal point scores. The absence of evidence on critical tradeoffs (e.g., stability, schedule, uncertainty) renders all models equally viable in a vacuum. Given the requirement for centralized policy management, real-time rule operations, and continuous monitoring across diverse campus network segments, Agile supports iterative development, rapid feedback loops, and responsive adaptation—key to managing dynamic security needs. However, without explicit evidence on legacy systems, verification cadence, or regulatory compliance, this decision remains provisional. The existing requirements emphasize operational control, auditability, and real-time visibility, which align with Agile’s emphasis on incremental delivery and stakeholder collaboration. Nevertheless, the unknown factors such as schedule, uncertainty, and regulation prevent a definitive conclusion; thus, the work groups must be structured to accommodate iterative progress while ensuring no requirement duplication, per prior validation failure.

Tradeoffs

1. Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.

Phases

Name: Requirements and verification planning

Phase type: planning

Activities

1. Review and baseline the source-defined work group: FR-001: The system shall manage firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system.; FR-002: The system shall allow authorized network administrators to create firewall rules.; FR-003: The system shall allow authorized network administrators to review firewall rules.; FR-004: The system shall allow authorized network administrators to update firewall rules.; FR-005: The system shall allow authorized network administrators to revoke firewall rules.; FR-006: The system shall enforce access restrictions between network segments.; FR-007: The system shall monitor blocked and allowed traffic.; FR-008: The system shall generate alerts for suspicious or unauthorized communication attempts.; FR-009: The system shall maintain audit logs of firewall policy changes and administrative actions.; FR-010: The system shall provide a dashboard for reviewing network security events.; FR-011: The system shall support multiple campus buildings.
2. Review and baseline the source-defined work group: NFR-001: The system shall be secure.; NFR-002: The system shall be scalable.; NFR-003: The system shall be reliable.; NFR-004: The system shall be manageable.
3. Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Versioned requirements baseline

2. Unit, integration, system and acceptance test plan
3. Unresolved-decision record

Entry criteria

1. Authoritative inputs and canonical requirements are available for review.

Exit criteria

1. A human reviewer has recorded the baseline decision and disposition of unresolved scope.

Security checks

1. Review implementation and acceptance evidence against FR-001: The system shall manage firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system.
2. Review implementation and acceptance evidence against FR-002: The system shall allow authorized network administrators to create firewall rules.
3. Review implementation and acceptance evidence against FR-003: The system shall allow authorized network administrators to review firewall rules.
4. Review implementation and acceptance evidence against FR-004: The system shall allow authorized network administrators to update firewall rules.
5. Review implementation and acceptance evidence against FR-005: The system shall allow authorized network administrators to revoke firewall rules.
6. Review implementation and acceptance evidence against FR-006: The system shall enforce access restrictions between network segments.
7. Review implementation and acceptance evidence against FR-008: The system shall generate alerts for suspicious or unauthorized communication attempts.
8. Review implementation and acceptance evidence against FR-009: The system shall maintain audit logs of firewall policy changes and administrative actions.
9. Review implementation and acceptance evidence against NFR-001: The system shall be secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Name: Agile delivery cycle

Phase type: iteration

Activities

1. Review and prioritize the backlog for the next iteration without changing source-defined scope.
2. Implement the source-defined work group: FR-001: The system shall manage firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system.; FR-002: The system shall allow authorized network administrators to create firewall rules.; FR-003: The system shall allow authorized network administrators to review firewall rules.; FR-004: The system shall allow authorized network administrators to update firewall rules.; FR-005: The system shall allow authorized network administrators to revoke firewall rules.; FR-006: The system shall enforce access restrictions between network segments.; FR-007: The system shall monitor blocked and allowed traffic.; FR-008: The system shall generate alerts for suspicious or unauthorized communication attempts.; FR-009: The system shall maintain audit logs of firewall policy changes and administrative actions.; FR-010: The system shall provide a dashboard for reviewing network

security events.; FR-011: The system shall support multiple campus buildings.

3. Implement the source-defined work group: NFR-001: The system shall be secure.; NFR-002: The system shall be scalable.; NFR-003: The system shall be reliable.; NFR-004: The system shall be manageable.
4. In an authorized isolated test environment, execute TEST-001: Given: the system manages firewall policies across academic, administrative, research, hostel, and guest networks When: a policy is updated via the centralized management interface Expected: the change is reflected across all network segments without service disruption
5. In an authorized isolated test environment, execute TEST-002: Given: authorized network administrators create firewall rules When: the actor initiates the creation of new firewall rules Expected: the system returns a newly created rule with specified parameters and status
6. In an authorized isolated test environment, execute TEST-003: Given: authorized network administrators request to review firewall rules When: the actor initiates a rule review operation Expected: the requested firewall rules are returned in full detail for examination
7. In an authorized isolated test environment, execute TEST-004: Given: authorized network administrators update existing firewall rules When: the actor initiates an update to a specific rule Expected: the system applies the updated rule and confirms its new state in the configuration
8. In an authorized isolated test environment, execute TEST-005: Given: authorized network administrators revoke firewall rules When: the actor initiates a rule revocation action Expected: the system removes the rule from active configuration and marks it as revoked
9. In an authorized isolated test environment, execute TEST-006: Given: the system enforces access restrictions between network segments When: a communication attempt is made between isolated segments Expected: the system blocks the connection.
10. In an authorized isolated test environment, execute TEST-007: Given: the system monitors traffic between network segments When: traffic flows through the firewall interface Expected: The system reports an observable result showing traffic flow states between network segments as monitored through the firewall interface.
11. In an authorized isolated test environment, execute TEST-008: Given: a suspicious communication attempt is detected When: an unauthorized or anomalous connection is initiated Expected: the system generates an alert indicating the event and its context
12. In an authorized isolated test environment, execute TEST-009: Given: system maintains audit logs of firewall policy changes and administrative actions When: the system records a policy change or administrative action Expected: The system records a policy change or administrative action and the external test captures the resulting response.
13. In an authorized isolated test environment, execute TEST-010: Given: system provides a dashboard for reviewing network security events When: an administrator accesses the dashboard interface Expected: the dashboard displays security event summaries with filtering and sorting capabilities
14. In an authorized isolated test environment, execute TEST-011: Given: system supports multiple campus buildings When: an administrative user initiates a network configuration request from each building's designated location Expected: the request is processed successfully in each location with no error or timeout
15. In an authorized isolated test environment, execute TEST-012: Given: A draft acceptance plan for the system to be secure When: An analyst reviews that acceptance plan Expected: The plan's measurable targets remain TBD or unconfirmed
16. In an authorized isolated test environment, execute TEST-013: Given: A draft acceptance plan for the system to be scalable When: An analyst reviews that acceptance plan Expected: The plan's measurable targets remain TBD or unconfirmed

17. In an authorized isolated test environment, execute TEST-014: Given: A draft acceptance plan for the system to be reliable When: An analyst reviews that acceptance plan Expected: The plan's measurable targets remain TBD or unconfirmed
18. In an authorized isolated test environment, execute TEST-015: Given: A draft acceptance plan for the system to be manageable When: An analyst reviews that acceptance plan Expected: The plan's measurable targets remain TBD or unconfirmed
19. Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
20. Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirement-linked implementation increment
2. Recorded test evidence
3. Reviewed feedback and next-iteration backlog

Entry criteria

1. Prior work, source scope and the reviewed test plan are available for iteration planning.

Exit criteria

1. A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.

Security checks

1. Review implementation and acceptance evidence against FR-001: The system shall manage firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system.
2. Review implementation and acceptance evidence against FR-002: The system shall allow authorized network administrators to create firewall rules.
3. Review implementation and acceptance evidence against FR-003: The system shall allow authorized network administrators to review firewall rules.
4. Review implementation and acceptance evidence against FR-004: The system shall allow authorized network administrators to update firewall rules.
5. Review implementation and acceptance evidence against FR-005: The system shall allow authorized network administrators to revoke firewall rules.
6. Review implementation and acceptance evidence against FR-006: The system shall enforce access restrictions between network segments.
7. Review implementation and acceptance evidence against FR-008: The system shall generate alerts for suspicious or unauthorized communication attempts.
8. Review implementation and acceptance evidence against FR-009: The system shall maintain audit logs of firewall policy changes and administrative actions.
9. Review implementation and acceptance evidence against NFR-001: The system shall be secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Name: Release and handoff

Phase type: release

Activities

1. Review the exact requirements baseline, implementation links and verification evidence before a release decision.
2. Carry unresolved source decisions into handoff; generated text is not human approval.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirements and verification evidence package
2. Pending human release decision
3. Outstanding-decision record

Entry criteria

1. Verification evidence and unresolved decisions are available for human review.

Exit criteria

1. A human has recorded the release/handoff decision for the reviewed content.

Security checks

1. Review implementation and acceptance evidence against FR-001: The system shall manage firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system.
2. Review implementation and acceptance evidence against FR-002: The system shall allow authorized network administrators to create firewall rules.
3. Review implementation and acceptance evidence against FR-003: The system shall allow authorized network administrators to review firewall rules.
4. Review implementation and acceptance evidence against FR-004: The system shall allow authorized network administrators to update firewall rules.
5. Review implementation and acceptance evidence against FR-005: The system shall allow authorized network administrators to revoke firewall rules.
6. Review implementation and acceptance evidence against FR-006: The system shall enforce access restrictions between network segments.
7. Review implementation and acceptance evidence against FR-008: The system shall generate alerts for suspicious or unauthorized communication attempts.
8. Review implementation and acceptance evidence against FR-009: The system shall maintain audit logs of firewall policy changes and administrative actions.
9. Review implementation and acceptance evidence against NFR-001: The system shall be secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Unknown factors: budget; delivery; failure_consequences; feedback; legacy; regulation; schedule; stability; uncertainty; verification

SDLC rankings and decision evidence

Policy: sequential-sdlc-1.1

Generation profile: sequential-sdlc-generation-2.0

Ranked candidates

Model: Agile

Score: 40

Reasons: No differentiating input evidence

Model: DevSecOps

Score: 40

Reasons: No differentiating input evidence

Model: Spiral

Score: 40

Reasons: No differentiating input evidence

Model: V-Model

Score: 40

Reasons: No differentiating input evidence

Model: Waterfall

Score: 40

Reasons: No differentiating input evidence

Model: Hybrid

Score: 40

Reasons: No differentiating input evidence

Model: Incremental

Score: 40

Reasons: No differentiating input evidence

Evidence register

Id: E-001

Source: description

Quote: Design a centralized firewall management system for a university that manages firewall policies across academic, administrative, research, hostel, and guest networks.

Id: E-002

Source: description

Quote: The system should allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts.

Id: E-003

Source: description

Quote: The platform should maintain audit logs of firewall policy changes and administrative actions, provide a dashboard for reviewing network security events, and support multiple campus buildings.

Id: E-004

Source: description

Quote: The solution should be secure, scalable, reliable, and manageable.

Source references

kb-c1e0239af9dd46bd:1 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 1

Page or section: text chunk 2

Relevance score: 0.7085

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:8 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 8

Page or section: text chunk 9

Relevance score: 0.7132

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kb-788cd8fba7ce0acb:3 - 03_OWASP_CheatSheets/083_OWASP_Network_Segmentation_Cheat_Sheet.md

Source document id: kb-788cd8fba7ce0acb

Chunk index: 3

Page or section: text chunk 4

Relevance score: 0.7036

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kb-c1e0239af9dd46bd:14 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 14

Page or section: text chunk 15

Relevance score: 0.7072

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:9 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 9

Page or section: text chunk 10

Relevance score: 0.6963

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

Validation findings

Code: human_semantic_review_required

Severity: warning

Section: document

Message: Review completeness, actor permissions, contradictions and citation applicability; schema and link checks do not prove semantic correctness.

Human review history

None recorded.

AI drafting record

Requested model: cybersrs-qwen3-4b-ft:latest

Actual models used: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Section models

Introduction: cybersrs-qwen3-4b-ft:latest

Functional requirements: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Non-functional requirements: qwen3:4b-instruct-2507-q4_K_M

Compliance and policy applicability matrix: cybersrs-qwen3-4b-ft:latest

Architecture: cybersrs-qwen3-4b-ft:latest

Threats: cybersrs-qwen3-4b-ft:latest

Risks: cybersrs-qwen3-4b-ft:latest

User stories and use cases: cybersrs-qwen3-4b-ft:latest

Verification and acceptance: cybersrs-qwen3-4b-ft:latest

Glossary: cybersrs-qwen3-4b-ft:latest

SDLC recommendation and workflow: qwen3:4b-instruct-2507-q4_K_M

Prompt version: sequential-1.33

Requirements traceability matrix

Requirement	Evidence / sources	Stories / tests / SDLC
FR-001	E-001	TEST-001, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-002	E-002	UC-001, TEST-002, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-003	E-002	UC-002, TEST-003, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-004	E-002	UC-003, TEST-004, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-005	E-002	UC-004, TEST-005, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-006	E-002	TEST-006, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-007	E-002	TEST-007, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-008	E-002	TEST-008, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-009	E-003	TEST-009, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-010	E-003	TEST-010, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-011	E-003	TEST-011, Requirements and verification planning, Agile delivery cycle, Release and handoff
NFR-001	E-004	TEST-012, Requirements and verification planning, Agile delivery cycle, Release and handoff
NFR-002	E-004	TEST-013, Requirements and verification planning, Agile delivery cycle, Release and handoff
NFR-003	E-004	TEST-014, Requirements and verification planning, Agile delivery cycle, Release and handoff

Requirement	Evidence / sources	Stories / tests / SDLC
NFR-004	E-004	TEST-015, Requirements and verification planning, Agile delivery cycle, Release and handoff

Model: cybersrs-qwen3-4b-ft:latest + qwen3:4b-instruct-2507-q4_K_M; prompt: sequential-1.33